

**Javier Eduardo
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Professional Profile

Mechanical engineer graduate from Universidad de los Andes, with a strong academic background in thermofluids and renewable energy systems, and complementary knowledge of combustion processes, aerodynamics, and the design and sizing of heat exchangers. Hands-on experience applying fundamental machine learning techniques and data processing methods to solve engineering problems. Proficient in finite element software for structural and fluid dynamics simulations, as well as CAD tools such as Fusion 360 and Autodesk Inventor. Skilled at developing technical reports backed by robust numerical analysis. Adaptable, analytical, and collaborative team player with strong communication skills, able to thrive in interdisciplinary environments and tackle challenges through a critical thinking approach.

Education

Mechanical Engineering Degree **Universidad de los Andes**

Bogotá, 2025

Relevant Coursework: Thermofluids III; Modeling and Numerical Analysis; Machine Design; CFD – Computational Fluid Dynamics; Machine Learning Applications in Remote Sensing

Academic Projects & Achievements

- Academic Exchange Program. *Académie Ste.Cécile International School*. Windsor, Canadá. April 24 – May 23, 2014
- Redesign Course. Universidad de los Andes, Bogotá. December 2023.
As a team, we redesigned the support structures for the gear motors driving the conveyor belts in the post-harvest line of the flower company Florval S.A.S. The goal was to extend the service life of the transmission shafts.
- Individual Project – Capstone Thesis. Universidad de los Andes, Bogotá. February 2024.
Conducted simulation and FEA analysis of a water-air heat exchanger for cooling applications in tropical climates using Ansys Fluent and Cadence Pointwise.
- Cumulative GPA (2025-01): 4.34/5.0

Work Experience

Volunteer Teacher **Fundación con las Manos – Bogota, Colombia**

February 2024 – April 2024

Planned and conducted semi-personalized and engaging academic tutoring sessions aimed at developing mathematical and socio-emotional skills in 5th grade students at Colegio La Giralda.

Maintenance Intern **Alimentos SAS – Bogota, Colombia**

February 2025 – July 2025

Collected data about the condition and performance of machines and facilities to verify proper operation. Contributed technical solutions to operational issues under supervision of senior engineers and supported the development of an application for spare parts inventory management.

Additional Training

- Basics of Air Conditioning & Heat Load Calculation–Certificate Coursera. L&T EduTech. August 20, 2025.
- Machine Learning Applications in Remote Sensing – Certificate of participation. Universidad de los Andes – Educación Continua. June 11, 2025
- Excel Fundamental for Data Analysis – Certificate Coursera. Macquarie University. February 4, 2025
- Introduction to Control System Analysis – Certificate Coursera. Universidad de los Andes. November 16, 2023

Relevant Skills

Languages: **Inglés** B2: Level 10-Academic English Support Program (AESP) [Uniandes]

Systems: **Programming Languages:** Matlab, Python, Excel (Intermediate Level)
Engineering Design Software: Fusion360, Autodesk Inventor, Ansys Workbench [Fluent & Mechanical]
Core Engineering Courses: Thermofluids III, Machine Design, Basic Aerodynamics, Combustion, CFD – Computational Fluid Dynamics
Supplementary Training: Investment Decision Analysis, Signals, Control Systems Analysis